National University of Computer and Emerging Sciences

Lahore Campus

Professional Practices in IT (CS 4001) (ALL Sections Except BCS-6C)

Date: 8th April, 2026

Course Instructor(s):

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Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 53

Total Questions: 5

Please answer all questions on question paper except question no. 2. Think before you write and write to the point. Read instructions for number of lines allowed for answers against each question. Do not write carelessly with bad hand writing.

Roll No ____________ Section ____________ Student Signature ____________

Question1: Syniad Software Ltd is preparing its year-end financial statements. The company has recently shifted from providing simple consultancy to developing a bespoke enterprise resource planning (ERP) system for a major client. This shift has resulted in a significant amount of "Work in Progress" and a change in their debt structure. **[9+2]**
[Answer on question paper]

1.1 Classification: From the list above, identify and group the items into the following four categories: **Fixed Assets**, **Current Assets**, **Current Liabilities**, and **Other Liabilities**.

1.2 Calculation: Calculate the **Total Working Capital** (Net Current Assets) for Syniad Software Ltd. Show your formula and all intermediate steps.

Financial Data as of December 31st:

Item	Amount (£)	Category from group	Working Capital
Computers and Servers (Net of depreciation)	£120,000	Fixed Asset	
Office Furniture and Fittings	£35,000	Fixed Asset	
Cash at Bank and In-Hand	£45,000	Current Asset	
Trade Debtors (Unpaid Client Invoices)	£180,000	Current Asset	
Work in Progress (Unbilled Project Hours)	£90,000	Current Asset	
Trade Creditors (Unpaid Supplier Invoices)	£75,000	Current liabilities	
Short-term Bank Overdraft	£40,000	Current liability	
Accrued Expenses (Unpaid Utilities & Rent)	£15,000	Current liability	
Long-term Bank Loan (Due in 2029)	£200,000	Other/long-term liability	

- **Total Current Assets:** £45,000 + £180,000 + £90,000 = **£315,000**
- **Total Current Liabilities:** £75,000 + £40,000 + £15,000 = **£130,000**
- **Working Capital Formula:** Current Assets – Current Liabilities

🎬 **Result:** £315,000 - £130,000 = **£185,000**

Question2: Recall the case study of Syniad Software L.td discussed during classes. The short introduction of the company to recall your memory is given. **Syniad Software Ltd** is a mid-sized software house founded by four directors who originally worked as developers. The company is in transition into growing from a small, informal team of 50 people into a larger organization with over 300 employees. The firm is structured into **Functional Departments** (Financial, Operations, Sales and marketing, Technical, Overseas), but it operates primarily on a project-by-project basis. **[Do not write more than 2 lines as answer for each following question] [2x10]**

2.1	***Q2.1: State two specific reasons why forcing a specialist to work abroad against their will leads to "long-term hurt" for Syniad's staff retention.*** 1. Potential loss of a highly specialized asset (resignation). 2. Damage to long-term morale/loyalty by ignoring personal life/aspirations.
2.2	***Q2.2: Briefly explain how under-capitalization limits Syniad's ability to "carry" specialists on the payroll when their specific skills do not match current billable contracts.*** The company cannot afford to keep "unproductive" specialists on the payroll; they must be billable to cover high overheads.
2.3	***Q2.3: Identify the primary factor that causes a specialist's group identity to become "diluted" when they are assigned to more than one project at a time.*** The specialist spends more time with the client's team/culture than their own "home" expert group.
2.4	***Q2.4: Why is the distinction between Centralization and Decentralization considered "academic" rather than practical for a company of Syniad's current size?*** In a small firm, directors are involved in everything; the formal "reporting lines" of a large PLC haven't formed yet.
2.5	***Q2.5: State the "fundamental principle" of quality control that makes the separation of Quality Management from day-to-day operations a necessity.*** "No one should be the judge of their own work." (Independence of quality).

2.6	***Q2.6: Contrast the conflicting priorities of the Operations Director (focused on billable hours) and the Technical Director (focused on R&D) regarding the allocation of technical staff.*** Operations wants staff on **Billable Projects** (Revenue); Technical wants staff on **R&D/Training** (Long-term capability).
2.7	***Q2.7: List advantage of the formal 6-month Staff Appraisal system replaces the "informal trust" and personal relationships found in the company's original 50-person core.*** 1. Provides a transparent record of achievements. 2. Standardizes promotion/reward criteria across the firm.
2.8	***Q2.8: Identify the two "perceived difficult problems" Syniad faces in maintaining corporate standards and loyalty as it expands into geographically dispersed regions.*** 1. Maintaining technical standards across distances. 2. Dilution of "corporate culture" in remote branches.
2.9	***Q2.9: In one sentence each, identify the primary financial risk of "building from scratch" versus the cultural risk of "buying an existing company" in a new market.*** Build: High cost/time and high financial risk. Buy: Cultural clash and risk of staff leaving post-acquisition.
2.10	***Q2.10: Which of the two expansion methods is described as "less risky but expensive," and which carries the specific risk of "destroying the value of the investment" if local staff resign?*** **Buying** is less risky/expensive in terms of market entry, but carries the risk of "destroying value" if staff resign.

Question3: From each of the following identify the most appropriate computer contract. Justify your answer in one line. Contracts list: A) **Custom built software at a fixed price, B)Contract hire, C)Times and materials, D) Consultancy [7x2]**

Scenario A The Compliance Portal : A regional government body, "Metro-Auth," requires a web portal to manage environmental permits. They have provided a **200-page functional specification** document that defines every user interaction. Due to strict public auditing, the budget is capped at **£150,000**, and the risk of cost overruns must remain entirely with the software house. Metro-Auth has no internal IT management to oversee the daily coding process. **Label: _A__, Justification:** The detailed 200-page specification allows for accurate quoting, and the client wants to transfer all financial risk to the supplier.
Scenario B: The Surge Requirement: "Global-Bank" is mid-way through developing its own mobile app. They have an experienced internal Project Manager and a clear technical roadmap. However, they have suddenly lost **three senior Java developers** to a competitor and are six months behind schedule. They do not want a third-party firm to take over the project; they simply need "bodies on seats" to work under the direction of their own Project Manager for a fixed six-month period. **Label: __B___, Justification:** The client retains all management control and project direction, simply "hiring" additional labor to work under their supervision.
Scenario C: The legacy maintenance crisis: "Vintage-Retail" operates a 15-year-old inventory system. The system is prone to random crashes, and the original documentation has been lost. The scope of repairs cannot be defined in advance because new bugs appear every week. "Vintage-Retail" wants a software house to provide ongoing support and bug-fixing, where they simply pay for the actual hours spent investigating and fixing issues as they arise. **Label: ___C__, Justification:** The unpredictable nature of bugs in undocumented code makes a fixed scope impossible; payment must be based on actual effort.
A regional hospital wants to modernize its patient records system. The IT department has produced a complete, exhaustive specification document detailing every interface, database schema, and security protocol required. The hospital board has a strict, locked budget allocated for this fiscal year and wants to guarantee that the final cost will not exceed a single, agreed-upon amount. **Label: __A_____, Justification:** The existence of exhaustive specifications allows the supplier to estimate costs accurately, meeting the client's requirement for a guaranteed price and a capped budget.
A financial technology company is planning to expand its mobile app to support cryptocurrency transactions. The internal development team is highly skilled in standard app development but has no experience with blockchain architecture or localized European crypto-regulations. They need a high-level subject matter expert for a period of four weeks to evaluate their current system design, assess legal risks, and provide a strategic roadmap before any coding begins. **Label: _D______, Justification:** The firm is seeking high-level expert advice, risk assessment, and strategic guidance rather than the delivery of a programmed software product or staff augmentation.
An established e-commerce firm is currently developing a new AI-driven recommendation engine in-house, managed by their own project director. Two of their senior back-end developers unexpectedly resigned to join a competitor. The firm urgently needs two competent developers to fill the gap for the next five months and work directly under the daily instructions of the internal project director to keep the build on track. **Label: __B____, Justification:** The client maintains daily management and project direction, requiring only the provision of specific human resources to fill gaps within their existing internal team.

You own a small software house. A potential client wants to build a mobile ordering app for their restaurant chain. During the first meeting, the client says: "We have a rough idea of what we want, but we will figure out the specific details as we go. We don't want any surprises on price. We want to pay a fixed price of $50,000 for the whole project. You handle everything." You own a small software house. A potential client wants to build a mobile ordering app for their restaurant chain. During the first meeting, the client says: "We have a rough idea of what we want, but we will figure out the specific details as we go. We don't want any surprises on price. We want to pay a fixed price of $50,000 for the whole project. You handle everything." **Label: __C___, Justification:** Because the requirements are poorly defined and the scope is expected to evolve, a fixed-price contract would pose an unacceptable financial risk to the software house.

Question4: From the following, identify each as either analytical or non-analytical job evaluation scheme. **[3 Marks]**

A startup called "Cloud-Scale" has grown from 5 to 50 employees in six months. To establish a quick salary structure, the CEO sits down with the Department Heads. They look at a list of all current roles (e.g., Junior Dev, Senior Dev, Office Manager, Sales Lead) and rank them from "Most Important to the Company" to "Least Important." No specific scorecards or skill matrices are used; the leaders simply compare the overall value of one role against another to place them in a hierarchy. **Answer: Non Analytical**
An established IT firm, "Data-Safe Corp," is conducting a formal audit to ensure fair pay across different departments. They use a standardized manual that requires every role to be scored on four specific factors: **Technical Complexity**, **Communication Requirements**, **Accountability for Budget**, and **Years of Specialized Experience**. Each factor is assigned a numerical value, and the total "points" for a Database Administrator are compared against the "points" for a Project Manager to determine their respective pay grades. **Answer:___ Analytical ______________**
A government-funded research institute is hiring for a "Lead Systems Architect" position. Before the job was even advertised, the HR department looked at the organization's pre-defined **Grade 8** characteristic description, which includes "Strategic Leadership" and "High-Level Technical Oversight." They decided the Architect role fits perfectly into Grade 8. They did not analyze the individual's specific coding languages or micro-skills but instead matched the general nature of the job to a pre-existing "Band" or "Grade." **Answer:___ Non Analytical __________________**

Question5: A startup is evaluating whether to build a B2B SaaS platform. Year 0 (current year) and year1 show investments, the remaining years show revenue. The projected cash flows are: **[2+2+1]**

Year Cash Flow (Rs.)	**Write Year Discount Factor**	Note: Year 0 is the initial investment. Year 1 is also a net outflow due to ongoing setup costs. Discount rate: 10%
0 (500,000) 1 (150,000) 2 180,000 3 280,000 4 320,000 5 350,000	0 1.0000 1 0.9091 2 0.8264 3 0.7513 4 0.6830 5 0.6209	

5.1 Calculate the NPV of this project. Should the startup proceed?

Answer:

Formula: Discount Factor (DF) = 1 / (1 + r)^t, where r = 10%. DCF = Cash Flow × DF. NPV = Σ DCFs.

Year	**Cash Flow (Rs.)**	**Discount Factor = 1/(1.10)^t**	**Discounted Cash Flow (Rs.)**
0	(5,00,000)	1 / $(1.10)^0$ = 1.0000	(5,00,000.00)
1	(1,50,000)	1 / $(1.10)^1$ = 0.9091	(1,36,363.64)
2	1,80,000	1 / $(1.10)^2$ = 0.8264	+1,48,760.33

3	2,80,000	1 / $(1.10)^3$ = 0.7513	+2,10,368.14
4	3,20,000	1 / $(1.10)^4$ = 0.6830	+2,18,564.31
5	3,50,000	1 / $(1.10)^5$ = 0.6209	+2,17,322.46

NPV calculation:
NPV = −5,00,000.00 − 1,36,363.64 + 1,48,760.33 + 2,10,368.14 + 2,18,564.31 + 2,17,322.46
NPV = Rs. 1,58,651.61 (POSITIVE)
Decision: Since NPV > 0, the project earns more than the required 10% return and creates value for the firm. YES — the startup should proceed.

5.2 Calculate the Payback Period.
Answer:
Payback Period = time needed for cumulative cash inflows to fully recover the initial investment (using UNDISCOUNTED cash flows).

Year	**Cash Flow (Rs.)**	**Cumulative (Rs.)**	**Status**
0	(5,00,000)	(5,00,000)	Not recovered
1	(1,50,000)	(6,50,000)	Not recovered
2	+1,80,000	(4,70,000)	Not recovered
3	+2,80,000	(1,90,000)	Not recovered (1,90,000 left)
4	+3,20,000	+1,30,000	**RECOVERED during Year 4**
5	+3,50,000	+4,80,000	Surplus

Fractional year calculation:
Amount still to recover at start of Year 4 = Rs. 1,90,000
Year 4 cash inflow = Rs. 3,20,000
Fraction of Year 4 required = 1,90,000 ÷ 3,20,000 = 0.59
Payback Period = 3 full years + 0.59 ≈ 3.59 years (≈ 3 years, 7 months)

5.3 A second project offers an NPV of Rs. 95,000 but a payback period of 6 years. The startup can only fund one. Which do you recommend and under what circumstances might your answer change?
Answer:
Comparison of the two projects against the standard criteria:

Criterion	**Project 1 (this project)**	**Project 2**	**Winner**
NPV	Rs. 1,58,652	Rs. 95,000	**Project 1**
Payback Period	3.59 years	6 years	**Project 1**
Value creation	Higher (more NPV)	Lower	**Project 1**
Capital recovery speed	Faster (3.59 yrs)	Slower (6 yrs)	**Project 1**

Recommendation: Choose Project 1. It dominates Project 2 on BOTH the NPV rule and the Payback rule — higher value creation AND faster capital recovery.
Circumstances under which the recommendation could change:

- RISK: If Project 1's cash flows are materially riskier or less certain than Project 2's (e.g., aggressive revenue assumptions), a smaller but more reliable NPV from Project 2 may be preferred.
- STRATEGIC VALUE: If Project 2 carries strategic upside beyond Year 5 (market entry, network effects, regulatory moat) that the 5-year NPV does not capture, it may still be preferred.
- COST OF CAPITAL: If the true discount rate is much higher than 10%, Project 1's NPV could compress more than Project 2's depending on cash-flow timing.
- FINANCING: If Project 2 has tied external financing (grant, strategic investor, co-funding) that Project 1 lacks, its effective cost of capital is lower and could flip the decision.
- LIQUIDITY: If the startup's cash is tight, Project 1's shorter payback is even more critical; if liquidity is ample, the longer payback of Project 2 is not a dealbreaker.